

RE for ML Intensive Contexts

Thank you for participating to this survey, in which we aim at collecting your opinion on requirements engineering for machine learning systems.

About data:

- Your answers will be anonymous;*
- We will collect social and demographical data in an anonymized form;*
- Collected data is going to be used for research purposes;*

Thanks in advance for your effort and time!

** Indica una domanda obbligatoria*

1. Email *

About the Study

NLP for requirement engineering is the use of natural languages processing techniques to analyze software requirements and extract relevant information such as entities, keywords, and sentiments. This can lead to more accurate and complete requirements specifications, improved system quality, and reduced development time and cost. By applying NLP techniques, stakeholders can better understand and manage requirements, resulting in improved system development outcomes. NLP for requirement engineering is a promising area of research and development that has the potential to improve the efficiency and effectiveness of system development processes.

Experiment Description:

We are conducting a survey to gather requirements for developing machine learning-intensive systems for a specific use case. As part of this activity, we will provide you with a problem statement related to the use case for which we are seeking requirements. In this survey, we are interested in understanding the specific needs and challenges of potential users in order to develop a system that is tailored to their requirements. To achieve this, we will be asking you (by open-ended questions) to emulate the behaviour of a software engineer, generating a list of 10 high-level requirements that will guide the development of the system.

ATTENTION: The survey responses will be stored securely and confidentially.

The survey responses, you provided, will be grouped with other submissions and used for specific research objectives. Result will be generalized and discussed from a generic point of view, but your personal data will be anonymized and unpublished to ensure confidentiality.

By clicking on the "next" button, you provide your consent to the above conditions.

Best regards

About yourself

Background Information

2. How old are you?

Contrassegna solo un ovale.

☐ < 18

☐ 18 - 30

☐ 31 - 50

☐ > 50

☐ Prefer not to say

3. Which of the following most accurately describe(s) you?

Seleziona tutte le voci applicabili.

☐ Female

☐ Male

☐ Non-binary

☐ Transgender

☐ Intersex

☐ Prefer not to say

☐ Altro: _____

4. Where are you based?

Contrassegna solo un ovale.

- ☐ Asia
- ☐ Africa
- ☐ North America
- ☐ South America
- ☐ Antarctica
- ☐ Europe
- ☐ Australia
- ☐ Prefer not to say

5. If you cover a working position, or you worked in the past, in which one(s) of the following professional roles do you have experience?

Seleziona tutte le voci applicabili.

- ☐ Data Scientist
- ☐ Software Engineer
- ☐ Data Engineer
- ☐ Software Developer
- ☐ Manager / Project Manager
- ☐ Software Analyst
- ☐ Software Architect
- ☐ Prefer not to say
- ☐ Altro: _____

6. How many years of working experience do you have for your current role?

Contrassegna solo un ovale.

- ☐ 0
- ☐ 1-3
- ☐ 4-6
- ☐ 7-9
- ☐ 10+

7. Have you ever worked on artificial intelligence enabled systems or on systems that include a machine learning module? *

Contrassegna solo un ovale.

☐ Yes

☐ No

Experimental Section

According to your experience in the machine learning and (or) software quality fields, we ask you to provide us with **10** examples of ***machine learning-specific*** requirements concerning a hypothetical machine understanding intensive system design to solve the issue reported in ***the problem statement***.

The requirements must specify the following:

- **Who** is/are the *final user(s)* of the application to realize (considering the application context(s));
- **What** is/are the *specific machine learning task/tasks* necessary to realize the application;
- **What** are the *principal* (and additional, if any) *benefit(s)* of using the proposed solution.

Some tips:

- Requirement elicitation is generally an effort and timing-consuming activity. Defining functional requirements is crucial to understanding and correctly analyzing the application domain and the starting assets, e.g., a problem statement. For these reasons, any usage of supporting material, like online resources, is allowed;
- If you are confident with some requirements elicitation standards (IEEE, User Story, ...), you can use them; otherwise, feel free to describe your ideas as you want;
- Remember that machine learning applications generally are related to data and often require more than one machine learning module to be realized. For these reasons, possible requirements could be concerned with data preparation and different machine learning algorithm and practices;
- The requirement specification you'll produce could be used for other tasks related to the Software Quality Metrics course, e.g., measurement activities. For these reasons is crucial for you and us to produce an exhausting requirements specification, trying to cover all the aspects reported in the problem statement fully.

Requirements Elicitation Standard Templates

User Story

As a [user type], I want to [action], so that [benefit].

A user story is a simple and effective way of expressing a user's needs and expectations for a product or service. It helps to prioritize features and design elements that are essential to the user experience, ensuring that the final product meets the user's needs and is easy to use.

Pattern IEEE 830

[The system] shall/must [perform some action or function] [under certain conditions or constraints].

The "shall statement" pattern (IEEE requirements specification pattern) provides a clear and concise structure for specifying functional requirements and helps ensure that all relevant aspects of a requirement are addressed. It also enables easy verification of whether a requirement has been met, improving communication and understanding among stakeholders, identifying potential issues and ambiguities early in the development process, and facilitating traceability from requirements to design, implementation, and testing.

Problem Statement

Heart disease is a major cause of morbidity and mortality worldwide. Early detection of anomalies in cardiac signals can help in the timely diagnosis and treatment of heart diseases. Cardiac signals such as electrocardiogram (ECG) and echocardiogram (Echo) provide valuable information about the electrical and mechanical activities of the heart. However, the interpretation of these signals can be challenging due to their complex spatio-temporal patterns.

A specific issue in this domain is the detection of anomalies in ECG and Echo signals. The goal is to identify abnormal patterns in the signals that can indicate the presence of heart disease. However, the complexity and variability of these signals make it difficult to identify relevant features and patterns. Traditional statistical methods such as threshold-based anomaly detection or clustering may not be optimal for detecting anomalies in these complex spatio-temporal signals.

To address this issue, an ML-intensive application can be developed using anomaly detection and spatio-temporal process learning algorithms. The application can use unsupervised anomaly detection algorithms such as autoencoders or one-class support vector machines to detect anomalies in the ECG and Echo signals. The application can also use spatio-temporal process learning algorithms such as hidden Markov models or recurrent neural networks to learn the spatio-temporal patterns in the signals and detect anomalies based on deviations from these patterns.

The data necessary to perform this task will require access to large-scale ECG and Echo datasets such as the PhysioNet or PTB diagnostic databases. The application can leverage cloud-based computing platforms and big data analytics to handle the high dimensionality and variability of the cardiac signals and achieve scalable and efficient anomaly detection and spatio-temporal process learning.

Overall, an ML-intensive application that leverages anomaly detection and spatio-temporal process learning algorithms can help overcome the challenges of detecting anomalies in ECG and Echo signals, leading to improved early diagnosis and treatment of heart disease, and ultimately better patient outcomes.

8. Requirement 1 *

9. Requirement 2 *

10. Requirement 3 *

11. Requirement 4 *

12. Requirement 5 *

13. Requirement 6 *

14. Requirement 7 *

15. Requirement 8 *

16. Requirement 9 *

17. Requirement 10 *

Closing

Thank you for taking the time to participate in this survey and provide us with your valuable feedback. We appreciate your contribution to our research. Your submission has been successfully stored, and a copy has been sent to you via email for your records.

Once again, thank you for your time and input.

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